

Suraj Bodapati

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WORK EXPERIENCE

Nomura Securities International

March 2023 - Present

Senior Software Developer

New York City, U.S.A

- Architected and migrated **10 applications** (9 critical data flows and 1 regulatory system) from end-of-support TIBCO EMS and TIBCO Rendezvous to firm-mandated Solace and Apache Pulsar event streaming in **6 months**, contributing to the firm's DORA compliance. Stood up Solace and Pulsar infrastructure and led a phased rollout across 10 teams (parallel releases, staggered rollouts, DR and rollback testing) with **zero production disruption** while maintaining uninterrupted data-processing and regulatory-reporting SLAs, and created reusable migration patterns for future platform transformations.
- Architected and deployed an AI-powered, context-aware chatbot using RAG over Confluence, Jira and Takara data-dictionary sources with LLM APIs, resolving recurring client queries, supporting developers on application logic and infrastructure configuration, and auto-generating custom Java and Python API code from natural language; built automated resync pipelines to keep the RAG vector database current, delivering 24/7 first-line support and a **~30% improvement in team productivity**.
- Built MCP tools and a harness-agnostic agent framework for LLM-powered natural-language retrieval of Takara reference data across **~30 data domains**, replacing multi-step UI workflows and custom data-join code; adopted by **20+ global teams** across Quants, Risk, Finance, Compliance, Legal, Research, and Technology, with reusable integration across agent frameworks.
- Migrated passwords and encryption keys across **100+ Takara components** to CyberArk Conjur vault with a secure, latency-conscious design; built an Ansible deployment pipeline that detects changes in an AES-256-encrypted config, decrypts it to a read-protected file at deploy time, and uses the extracted key to fetch credentials at runtime.
- Developed and maintained Ansible automation for cybersecurity vulnerability remediation across Takara (removing vulnerable files and JARs, adjusting permissions, updating Java and out-of-date software), achieving **97% within-SLA remediation**.

Oracle

May 2022 - Aug 2022

Software Developer Intern, MySQL Heatwave Team

Buffalo, U.S.A (Remote)

- Designed and developed an asynchronous metrics-ingestion framework using AWS SQS and Lambda to decouple periodic MySQL HeatWave metric persistence from synchronous processing, **improving metric-processing throughput by ~20%**; implemented automated error handling, logging, retries, and SQS Dead Letter Queue-based reprocessing for reliable AWS-to-OCI usage and billing data transfer.
- Developed automated scripts using AWS CDK to create AWS SQS, Lambda resources and corresponding IAM roles during CI/CD process.
- Improved unit test coverage of AWS lambda functions from **72% to 94%** using pytest framework.

JP Morgan Chase & Co.

Jul 2019 - Jul 2021

Software Engineering Program, Corporate Technology (CT)

Hyderabad, India

- Built the Kafka-based microservice integration framework for InstaMed client onboarding across **7 systems** and **3 lines of business**, enabling asynchronous straight-through processing with Resiliency4J patterns, automated failed-request reprocessing and integrated log monitoring for auto recovery. Owned the majority of critical components while requirements changed up to go-live week, and delivered the CEO-visible project on time, earning direct appreciation from JPMC's global CIO.
- Developed core functionality and RESTful APIs for a key Client KYC and onboarding application used across JPMC with Java Spring Boot, Hibernate and Swagger, and built Angular UI components in parallel.
- Volunteered to build a ReactJS and Spring Boot application for the NGO Dr. Reddy Foundation (DRF), used to connect and maintain a network of **5,000+ DRF** alumni across India.

PROJECTS

National Science Foundation Research Assistant, 2022

Chatbot to Combat Disinformation and Scam: Developed an NSF-funded chatbot for a government senior-citizen training program that simulated legitimate government officials in real-time conversations to test participants' resistance to PII and financial-information scams; deployed across WhatsApp, Telegram, and Facebook Messenger, evaluating **80+ participants** across **200+ simulated scam interactions**, with failed assessments triggering mandatory retraining.

EDUCATION

M.S. in Computer Science, 3.76/4.0 **University at Buffalo (State University of New York)**

New York | August 2021 - December 2022

Bachelor of Engineering in Information Technology **Chaitanya Bharathi Institute of Technology (CBIT)**

Hyderabad, India | 2015 - 2019

AWARDS & HONORS

Nomura STAR and Nomura SPOT awards

Nomura 2023-2026

Execution Excellence Award, Q1 and Q2 2021 (firm level)

JPMC 2021

Lead Author of three research papers in international journals indexed by IEEE and Springer

SKILLS

Languages & Frontend: Java, Python, TypeScript, JavaScript, C++, SQL, Shell Script, HTML, CSS, Angular.js, React

Backend & Messaging: Spring Boot, REST APIs, gRPC, Swagger, Kafka, Solace, Apache Pulsar, TIBCO EMS/Rendezvous, Node.js, Flask

AI / LLM: LLM APIs, RAG, LangChain, LangGraph, Amazon Bedrock, Model Context Protocol (MCP), Agent Harness, NLTK, AWS Certified AI Practitioner

Cloud, DevOps & Testing: AWS (SQS, Lambda, CDK, IAM), Pivotal Cloud Foundry, Docker, Jenkins, Ansible, CI/CD, Git, Jira, JUnit 5, TestNG, pytest

Security & Observability: CyberArk Conjur, Sonar, Splunk, Grafana, RiverMuse, Elasticsearch